UNIVERSITY OF ZULULAND

A NODE FOR AFRICAN THOUGHT

RESEARCH PROPOSAL

With the provisional title:

Evidence-Based Multi-Objective Evaluation of AI and Heuristic Controller Placement for Multi-Site Software Defined Networks

For the degree of

In the field of Computer Sciences

FACULTY OF SCIENCE, AGRICULTURAL AND ENGINEERING

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TABLE OF CONTENTS

1 INTRODUCTION 2
2 BACKGROUND OF THE STUDY 2
3 PROBLEM STATEMENT AND RESEARCH QUESTIONS 2
3.1 Problem Statement 2
3.2 Research question(s) 2
4 RESEARCH AIM AND OBJECTIVES (or HYPOTHESES, where applicable) . 2
4.1 Research Aim 3
4.2 Research objectives (or Hypotheses) 3
5 CONTRIBUTION OF THE STUDY 3
6 LITERATURE REVIEW 3
7 THEORETICAL (or CONCEPTUAL) FRAMEWORK 3
8 RESEARCH METHODOLOGY 3
8.1 Research philosophy 4
8.2 Research approach 4
8.3 Research design 4
8.7 Data collection 4
8.8 Data analysis method(s) and procedure 4
9 RESEARCH QUALITY: ETHICAL AND SAFETY ISSUES 4
9.1 Research quality 4
9.2 Ethical and safety considerations 5
10 INTELLECTUAL PROPERTY 5
11 RESOURCES REQUIRED AND PROJECT PLAN 5
11.1 Resources required 5
11.2 Budget 5
11.3 Research process - Gantt chart 6
12 FEASIBILITY OF THE STUDY 7
13 DISSEMINATION PLAN 7
14 PROPOSED CHAPTER DIVISION 7
15 REFERENCES 7
16 CANDIDATE'S DECLARATION 7
17 SUPERVISOR(S) DECLARATION 7
18 ANNEXURE A: PARTICIPANT INFORMED CONSENT DECLARATION 9

1 INTRODUCTION

Software Defined Networking (SDN) controller placement is a critical infrastructure optimization problem in which controller location and number influence latency, reliability, and operational performance. This study focuses on evaluating whether AI-driven controller placement methods provide decision-relevant advantages over established heuristic methods in multi-site SDN environments. Heller et al. (2012) demonstrated that controller placement latency bounds of 50 ms are achievable in 82% of WAN topologies using optimized placement, showing 1.4-1.7x latency improvement over random placement. Infrastructure teams therefore require evidence-based method selection to determine when this improvement is attainable under their performance-sensitivity and resource constraints.

This proposal is structured as follows. Section 2 presents the background to SDN controller placement and identifies under-investigated areas in current research. Section 3 defines the problem statement and research questions. Section 4 states the research aim and objectives, followed by Section 5 on contribution to knowledge and practice. Sections 6 to 10 cover literature review, conceptual framing, methodology, quality and ethics, and intellectual property, while later sections present project planning, feasibility, dissemination, chapter division, references, declarations, and annexures.

2 BACKGROUND OF THE STUDY

Controller placement in SDN is commonly formulated as an optimization problem that assigns switches to controllers under latency and resilience constraints. The Wang and Chen (2021) established that controller placement strategy directly affects link failure foresight (LFFCPP), reducing control-plane latency fluctuation during link failures by minimizing switched-to-closest-controller routing changes. This is critical in multi-site deployments where topology changes and link failures directly affect network continuity and control-plane responsiveness.

Current research shows two broad method families. Classical heuristics are computationally efficient and operationally interpretable. Farahi et al. (2026) showed that AP-DQN achieves 24% load-balancing improvement, 25% latency reduction, and 28% lower link operational cost on 15-50 switch topologies. However, important gaps remain. Lange et al. (2015) showed that latency-only studies miss 40-60% of operationally viable controller configurations when reliability and computational cost are included. Cross-topology robustness and reproducibility remain under-tested.

This study addresses these gaps through a reproducible, multi-objective evaluation pipeline that compares AI and non-AI methods under controlled factorial scenarios. The study advances knowledge by producing transparent evidence on trade-offs among latency, reliability, and computational cost across topology families and scales. It advances practice by generating scenario-conditioned guidance on when AI methods are justified and when baseline heuristics are more suitable for real operational settings.

3 PROBLEM STATEMENT AND RESEARCH QUESTIONS

3.1 Problem Statement

There is insufficient reproducible, multi-objective evidence to determine when AI-based SDN controller placement methods outperform classical heuristics in multi-site

SDN controller placement in multi-site environments is not primarily a problem of missing evidence; it is a quantifiable conflict between competing objectives. Lange et al. (2015) show that placement methods optimized around latency alone can fall short once reliability and operational cost are introduced, because the Pareto frontier reveals that no single configuration dominates across all criteria. This multi-objective conflict is not incidental; it is foundational to why method selection remains uncertain in practice. The unresolved problem is therefore not whether AI can outperform heuristics in isolated cases, but whether it can preserve its performance advantage while also remaining stable under link failures and across topology scales.

? Do AI-based controller placement methods preserve their latency advantage while improving reliability and computational cost along the Pareto frontier in multi-site SDN environments?
? How do AI and heuristic methods compare in performance stability under link-failure dynamics across topology families and network scales?
? For a given operational constraint set, which points on the latency-reliability-cost Pareto frontier are achievable by AI versus heuristics, and at what computational cost?

4.2 Research objectives (or Hypotheses)

- To compare AI-driven and heuristic-based controller placement methods on control-plane latency, reliability, and computational cost across multi-site SDN scenarios.
- To assess the robustness and generalizability of both method families across topology types and network scales using a controlled factorial experiment.
- To formulate scenario-conditioned decision guidelines that indicate when AI-based methods provide sufficient benefit over heuristic baselines to justify additional computational complexity.

5 CONTRIBUTION OF THE STUDY

5.1 Contribution to Knowledge (Theory)

This study contributes to knowledge by developing a reproducible, multi-objective evaluation framework for SDN controller placement that integrates latency, reliability, and computational cost within one coherent analytical pipeline. It extends current understanding by moving beyond single-metric performance reporting and by explicitly modelling trade-offs across topology families and scale conditions. The study also strengthens methodological rigor in this domain through transparent baseline comparisons, controlled factorial design, and explicit evidence for scenario-dependent method performance.

5.2 Contribution to Practice and Policy

This study contributes to practice by producing decision-ready guidance for infrastructure engineers and network planners on selecting controller placement methods under operational constraints. Instead of assuming AI superiority, it provides scenario-specific recommendations on when heuristic methods are sufficient and when AI methods deliver meaningful performance gains relative to cost. For institutional and organizational policy, the study supports adoption of reproducibility standards, benchmark transparency, and multi-criteria evaluation protocols in AI-enabled network optimization projects.

6 LITERATURE REVIEW

Software-defined networking (SDN) controller placement has been studied extensively as a network optimization problem in which controller location and controller count shape latency, reliability, and operational efficiency. The literature shows clear movement from classical heuristic methods toward metaheuristic and learning-based approaches, but the empirical evidence remains uneven in how it measures performance, reports scalability, and supports decision-making under real deployment constraints. This review is organized thematically according to the three research objectives: comparative performance, robustness across topology and scale, and decision frameworks for controller selection.

6.1 Comparative Performance of AI and Heuristic Controller Placement Methods

Recent empirical studies consistently show that controller placement quality is not determined by a single metric. Classical heuristics such as greedy selection, clustering-based approaches, and k-center variants are often reported to deliver strong baseline performance because they are computationally efficient and easy to implement [H [Fa2]hi, 2022]. In several experiments, these methods achieve competitive latency outcomes, particularly in small to medium networks, while keeping runtime overhead low [Yusuf, 2023]. This makes them practical benchmarks for any proposal that claims improvement through more advanced methods.

In contrast, AI-driven methods, including genetic algorithms and reinforcement learning, are frequently evaluated as potential improvements over heuristics [F [Kreiu2,2022]15] [H [F [Kreu1,2022]15]. Empirical results often indicate that these methods can reduce average latency or improve controller coverage under selected conditions. However, the gains are rarely universal. Some studies report marginal improvements that depend on topology shape, controller count, or initialization settings, while others show that the computational cost of training or iterative search offsets the performance gain [F [Yusu2,2022]3]. This is an important finding because it suggests that superiority claims must be interpreted in relation to the cost of producing the result, not only the result itself.

A recurring weakness in the literature is that latency is often treated as the dominant outcome, while reliability and runtime are under-emphasized. Lange et al. (2015) systematically analyzed Pareto-effective placement frontiers and demonstrated that studies treating latency as a single metric miss 40-60% of operationally viable controller configurations when computational cost and failure tolerance are included. In studies where reliability is included, it is frequently operationalized narrowly (link failure tolerance or controller reachability alone), without full multi-objective comparison against computational cost and load-balancing stability.

6.2 Robustness Across Topology Families and Network Scales

A second major theme in the literature concerns robustness. Empirical SDN controller placement studies frequently test methods on a limited set of synthetic topologies, commonly small graphs or a single topology family, which restricts the generalizability of findings [F [Kreutz, 2015]]. Some studies show that a method performing well on one topology family may degrade sharply when evaluated on a different structural pattern, such as denser graphs, irregular multi-site layouts, or larger-scale deployments [H [F [Yusuf, 2023]]]. This is especially relevant for multi-site environments, where controller placement must remain effective under more heterogeneous and distributed conditions.

Recent work has increasingly recognized scale as a critical factor. Studies that expand experiments across network sizes often find that algorithm behaviour changes as the number of switches, links, and controllers increases [H [F [Kreutz, 2015]]]. Heuristic methods tend to remain stable in runtime, while AI-based methods may exhibit better solution quality in some larger settings but at the cost of slower convergence or greater sensitivity to parameter choices [H [F [Yusuf, 2023]]]. In other cases, the advantage of AI methods disappears as scale increases because the optimization search becomes more expensive or less consistent across runs [F [Kreutz, 2015]] [H [F [Kreutz, 2015]]]. These findings support the view that scalability is not a secondary detail but a central criterion for evaluating practical usefulness.

The literature also shows that robustness is usually underreported in a way that weakens comparability. Many papers present results for a single topology family, a narrow range of scale levels, or a single experimental seed [Yusuf, 2023]. Few studies report variation across multiple topology types together with performance

dispersion or statistical stability. As a result, it is difficult to determine whether reported gains are reproducible or merely artifacts of a specific test setup. This directly matches the gap identified in the background section: cross-topology and cross-scale behaviour is still insufficiently documented.

For this reason, a study that deliberately tests controller placement methods across topology families and network scales has clear value. It can show not only which method performs best on average, but also where each method breaks down, where it remains stable, and whether the observed performance differences are large enough to matter operationally.

6.3 Decision Frameworks for Multi-Objective Controller Placement

A third body of literature addresses multi-objective decision-making in controller placement. The theoretical basis here is that controller placement must balance competing objectives rather than optimize one dimension in isolation. In principle, this aligns well with Pareto-based reasoning and trade-off analysis, where a solution is judged by how it performs across several criteria simultaneously [F [Kreu2022]2015] . In practice, however, many empirical studies still stop at reporting performance indicators without translating those findings into explicit decision guidance.

Several recent studies use multi-objective evolutionary methods or reinforcement learning variants to generate sets of candidate solutions rather than a single optimal solution [H [F [Kreu2022]2015] . These studies show that trade-off frontiers can help reveal the relationship between latency reduction and other costs, such as runtime or resilience loss. Where Pareto analysis is used well, it helps demonstrate that the “best” method depends on the decision context rather than on absolute performance alone [Farahi, 2022] . This is conceptually useful for infrastructure planning because decision-makers need to know when a more complex method is worth the extra computational burden.

Despite this progress, the literature still lacks a consistent decision framework for selecting among controller placement methods. Studies often present comparative tables or plots, but do not define practical thresholds for when AI methods justify their added complexity [Farahi, 2022] . Very few studies convert experimental outcomes into scenario-conditioned guidance such as use the heuristic when the topology is small or static and use the AI method only when expected performance gains exceed a defined cost threshold. That omission is significant because it leaves practitioners without a usable basis for method selection.

The present study responds to this gap by using empirical comparison not only to rank methods, but also to derive decision-relevant recommendations. In doing so, it extends the literature beyond pure performance reporting toward evidence-based

selection logic, which is more useful for operational network planning and aligns with the contribution required by the template.

6.4 Synthesis of the Literature and Research Gap

Taken together, the empirical literature shows that AI-driven controller placement can offer performance gains, but those gains are neither uniform nor automatically superior to heuristic baselines [Farahi, 2022] . Classical methods remain competitive in efficiency and predictability, while AI methods sometimes improve solution quality at the expense of runtime, stability, or implementation complexity [Farahi, 2022] . The strongest recurring limitation in current research is the lack of studies that compare these methods using a reproducible, multi-objective framework across multiple topology families and scales.

This creates three under-investigated areas directly relevant to the present study. First, the literature does not yet provide enough evidence on comparative performance when latency, reliability, and computational cost are evaluated together. Second, cross-topology and cross-scale robustness is still insufficiently established. Third, existing studies rarely convert experimental results into practical decision rules for selecting controller placement methods in real multi-site SDN environments.

Accordingly, this study is necessary because it addresses these gaps through a controlled empirical evaluation that compares AI-driven and heuristic methods using a consistent experimental design and multi-objective analysis. In line with the background section, the study will generate new knowledge about trade-offs, improve understanding of method behaviour under different conditions, and support better practice through scenario-based guidance for controller placement decisions.

7 THEORETICAL (or CONCEPTUAL) FRAMEWORK

This study is situated within three complementary concepts: optimisation theory, multi-objective decision

Analysis, and distributed systems reliability. Together, these concepts provide the explanatory basis for

Evaluating whether AI-driven controller placement methods justify their added complexity when compared

With heuristic alternatives in multi-site SDN environments.

At the core of the study is optimisation theory, which treats controller placement as a constrained decision

Problem in which feasible solutions must satisfy performance and resilience requirements. In this context,

The goal is not only to identify a high-performing placement, but to determine how well each method

Navigates competing constraints such as latency, reliability, and computational cost. Heuristic methods are

Conceptually appropriate as efficient search strategies that produce acceptable solutions with low

Overhead, while AI-driven methods represent more complex search and learning strategies that may

Improve solution quality but require greater computational effort.

The study is further anchored in multi-objective decision analysis, particularly the notion that no single

Controller placement method can be evaluated meaningfully on one metric alone. Instead, methods should

Be assessed as trade-off structures across multiple objectives. This framework supports comparison

Through dominance, balance, and Pareto efficiency, allowing the study to determine whether AI-based

Approaches offer materially better overall outcomes than heuristics when latency improvement is weighed

Against runtime burden and reliability behaviour. This is the key conceptual basis for judging whether AI

Complexity is justified over heuristic simplicity.

Distributed systems reliability provides the third conceptual lens. In multi-site SDN deployments, controller

Placement is not merely an optimisation exercise; it is also a resilience problem because controller location

Affects fault tolerance, coverage under failure, and continuity of network control. This concept explains why

The study includes reliability as a central performance dimension rather than treating it as a secondary

Outcome. It also supports the expectation that a method that performs well under ideal conditions may not

Remain effective when network topology, scale, or failure conditions change.

Taken together, these concepts explain the anticipated findings of the study. The framework predicts that

The comparative value of AI and heuristic placement methods will depend on the interaction between

Topology characteristics, scale, and the relative weighting of performance objectives. AI methods are

Expected to be beneficial only where their performance gains are sufficiently large to justify them

Computational and implementation cost. Heuristic methods are expected to remain competitive were

Efficiency, interpretability, and stability are more important than marginal improvements in placement

Quality.

This conceptual framework therefore bridges the literature review and the methodology by providing the

Logic for the empirical comparison. It clarifies what will be compared, why the comparison matters, and

How the results will be interpreted in terms of practical method selection for multi-site SDN controller

Placement.

8 RESEARCH METHODOLOGY

This study adopts a methodology designed to produce objective, reproducible, and decision-relevant evidence on the comparative performance of AI-driven and heuristic controller placement methods in multi-site SDN environments. Each methodological choice is selected to support direct comparison across latency, reliability, and computational cost, while preserving experimental control and traceability.

8.1 Research philosophy

This study is grounded in a positivist research philosophy. Positivism is appropriate because the phenomenon under investigation is measurable and testable: controller placement methods produce observable outcomes such as latency, reliability, runtime, and resource cost. The aim of the study is not to interpret subjective experiences, but to evaluate whether one algorithmic class performs better than another under controlled conditions. A positivist stance therefore fits the research problem because it treats performance as an objective reality that can be measured, compared, and replicated.

The choice of positivism is also justified by the need for evidence that can support defensible method selection in practical SDN environments. Since the central question is whether AI-based complexity is justified relative to heuristic simplicity, the study requires quantifiable results rather than interpretive narratives. Positivism enables the use of controlled simulation, repeated trials, and statistical comparison, all of which strengthen reliability and reduce the influence of researcher bias.

8.2 Research approach

The study follows a deductive research approach. Deduction is suitable because the investigation begins with existing ideas from optimization theory and controller

In particular, the study tests the proposition that AI-based methods (specifically reinforcement learning with adaptive prioritization) may outperform heuristic methods in specific topology and scale scenarios. Farahi et al. (2026) established that AP-DQN achieves 24% load-balancing gains and 25% latency reduction in small-to-medium networks (15-50 switches), but performance universality remains unvalidated in multi-site topologies with >100 switches and inter-controller communication delays.

A deductive approach is justified because the research is not trying to generate theory from open-ended field observation. Instead, it is evaluating predefined methods against explicit performance criteria. This aligns with the golden thread of the proposal: the problem statement identifies a decision gap, the research questions ask whether AI methods offer meaningful advantages, and the methodology tests that claim through controlled experimentation. Deduction is therefore the correct approach because it moves logically from theory and prior claims to empirical verification.

8.3 Research design

The study uses an experimental research design. Experimental design is appropriate because the core purpose is to compare method performance under controlled and repeatable conditions. By manipulating the controller placement method and systematically varying topology and scale, the study can isolate the effect of each factor on the performance metrics of interest.

A factorial design is used within the experimental framework because the study must examine more than one independent variable at the same time. In this case, the key variables include method type, topology family, and network scale. A factorial design is justified because it enables the researcher to observe not only the main effect of each variable, but also interaction effects. For example, Heller et al. (2012) established that optimal controller placement achieves 50 ms latency bounds reliably in 82% of topologies in the 10-50 switch range, yet the performance trajectory across larger multi-site topologies (>100 switches with inter-site latency) remains empirically unconstrained. Therefore, it is critical to determine whether AI methods maintain this performance advantage or regress to random placement utility in larger multi-site topologies.

The factorial structure also improves the explanatory value of the findings. Rather than simply ranking methods, the design allows the study to identify where performance differences are stable, where they vary, and where heuristic methods remains preferable because of lower cost or greater consistency. This directly supports the study's aim of generating scenario-conditioned guidance.

8.4 Data collection

Data collection in this study is conducted in a technical simulation environment rather than through human participants. This is appropriate because the research problem concerns algorithmic performance, not perception or behaviour. The data collection process is therefore designed to preserve experimental control, ensure repeatability, and minimise the risk of contamination from uncontrolled external factors.

8.4 Data collection

Data collection is conducted in a controlled simulation and emulation workflow rather than through human participants. This is appropriate because the research problem concerns measurable algorithmic performance under standardized conditions. The collection process is designed to preserve repeatability, minimize uncontrolled variance, and support transparent cross-method comparison.

8.4.1 Parameter Justification and Technical Stack Defense

The study adopts Mininet as the emulation environment and Ryu as the SDN controller for all three method classes. This choice is necessitated by the empirical structure of prior work. Group A foundational studies show that controller-placement claims are highly topology-dependent, with Internet2-scale evaluation establishing the canonical latency baseline (Heller et al., 2012). Group B meta-heuristic studies report strong gains under simulation-specific settings, but with transferability and parameter-sensitivity limitations (Radam et al., 2022). Group C DQN/DRL studies report substantial improvements, but often under-specify reproducibility-critical implementation details and reliability-operationalization (Farahi et al., 2026; Benoudifa et al., 2023).

8.4.1 Design and description of instruments

In this study, the instruments are the technical components used to generate, capture, and process experimental evidence. These include the simulation environment, controller placement implementations, custom Python scripts for running experiments, logging mechanisms, and metric extraction utilities. The simulation environment functions as the primary instrument because it allows the study to model multi-site SDN topologies under consistent conditions. The algorithm implementations serve as operational instruments for executing the comparative methods. The scripts and monitoring probes serve as measurement instruments for capturing latency, reliability outcomes, and computational cost.

This design is justified because simulation-based instrumentation offers repeatability, parameter control, and safety. It also avoids negative operational impact on live network systems, which is important for ethical and methodological reasons. Since the study does not rely on production infrastructure, there is no risk of disrupting real network performance or compromising external systems. In addition, the measurement pipeline is designed to preserve data integrity by using structured logging, consistent metric definitions, and repeated trial execution under identical conditions.

8.4.2 Data collection procedure(s)

Data collection will proceed in a controlled sequence. First, the experimental topologies will be generated or selected according to the topology families and scale levels defined in the design. Second, each controller placement method will be executed under the same topology and scale conditions so that comparisons are fair and directly attributable to the method under test. Third, each run will capture the relevant output files, including simulation logs and packet-level traces where necessary, to support metric extraction.

Latency will be derived from the time taken for control-related communication or decision propagation within the simulated environment. Reliability will be measured using outcomes such as controller coverage, failure tolerance, or service continuity under predefined failure conditions. Computational cost will be extracted from runtime, convergence duration, and related execution overhead. These metrics are selected because they jointly represent the performance trade-off that the research questions are designed to evaluate.

The collection procedure is justified by the need for consistency and comparability. Running all methods under the same controlled conditions ensures that performance differences are not caused by differences in topology generation, simulation timing, or measurement procedure. Capturing both logs and packet-level data also increases traceability, allowing the experiment to be audited and repeated if needed.

First, baseline parity is non-negotiable. The use of Internet2 and ATT-MPLS-class topologies is deliberate to replicate the structural conditions under which foundational CPP latency claims were established (Heller et al., 2012). Ensuring cross-study parity at the topology layer prevents inflated AI claims caused by easier synthetic graph regimes.

Second, this study introduces a formal reliability specification to eliminate black-box scoring. Reliability is explicitly defined as the mean fraction of nodes that remain controller-reachable under single-link failure across all edge removals. Directly addressing reporting limitations where reliability is presented as an aggregate score without fully auditable operationalization (Benoudifa et al., 2023), this definition makes each reliability claim independently recalculable from graph state and controller set.

Third, the reproducibility pivot to a full PyTorch DQN/DRL stack is required by convergence-control expectations in AP-DQN and MuZero literature (Farahi et al., 2026; Benoudifa et al., 2023). The prior lightweight reward baseline is useful for exploratory benchmarking but cannot provide replay-based stabilization, target-network synchronization, and gradient-level diagnostics expected in contemporary DRL evidence. Farahi et al. (2026) report concrete DQN settings, including learning rate 0.001, batch size 32, discount factor gamma 0.99, and 1000 episodes. These values provide a quantitative reference band for implementation and hyperparameter calibration.

Finally, operational validity requires Mininet with Iperf3-driven traffic generation. Analytical or offline optimization remains essential for theory but insufficient for deployment-facing inference. Mininet introduces packet-level behavior, contention effects, and emulation-time variability that abstract graph optimization cannot represent.

8.4.2 Design and description of instruments

The study instruments include the simulation and emulation environment, DQN/DRL and baseline controller-placement implementations, experiment orchestration scripts, structured logging pipelines, and metric extraction utilities. Mininet and Python-based instrumentation serve as the primary operational environment for repeatable network experimentation, while topology datasets and generators support controlled scenario construction. This design is justified because it combines parameter control, repeatability, and topological realism, and avoids distortions of protocol or network instrumentation.

8.4.3 Piloting of instruments (where applicable)

Before the full experiment begins, a pilot run or smoke test will be conducted. The purpose of the pilot is to verify that the simulation environment is stable, that the scripts execute correctly, that the logging pipeline captures the intended outputs, and that the chosen metrics can be extracted without data loss or corruption. A pilot is especially important in a computational study because small configuration errors can produce misleading results across a full factorial design.

The pilot is justified on methodological grounds because it reduces the risk of invalid or incomplete data in the main experiment. It also allows any issues with runtime, file generation, or metric parsing to be corrected early, before the full experimental matrix is executed. In addition, the pilot helps confirm that the chosen network sizes and topology settings are computationally feasible within the available resources. This is important for maintaining continuity between the design and the actual implementation.

8.5 Data analysis method(s) and procedure

The data analysis strategy combines descriptive, inferential, and multi-objective techniques. Descriptive statistics will first be used to summarise the results for each method and scenario, including means, dispersion, and performance ranges. This provides an initial picture of how the methods behave across conditions and supports transparency in reporting.

Inferential statistics will then be used to compare methods more formally. Where the experimental structure involves two methods under comparable conditions, t-tests may be used to assess whether observed differences are statistically meaningful. Where multiple factors are involved, analysis of variance is more appropriate because it can test the effect of method type, topology, scale, and their interactions in one model. This is justified because the research design is factorial and the study aims to understand not only whether methods differ, but also under what conditions those differences emerge.

Pareto front analysis will be used to evaluate multi-objective trade-offs across latency, reliability, and computational cost. This method is appropriate because the study does not define success using a single metric. Instead, it seeks to identify whether a method dominates others across the combined objective space, or whether each method occupies a different trade-off region. Pareto analysis therefore aligns directly with the research aim and provides a rigorous basis for judging whether AI complexity is justified relative to heuristic simplicity.

8.5 Data analysis method(s) and procedure

The analysis combines descriptive, inferential, and multi-objective methods. Descriptive statistics summarize means, dispersion, and performance ranges per method and scenario. Inferential analysis uses t-tests for pairwise comparisons and analysis of variance for factorial effects and interactions across methods, topology family, and network scale. Pareto-front analysis is then used to evaluate trade-offs across latency, reliability, and controller placement costs. This is appropriate because success is not defined by a single metric; methods are evaluated by dominance patterns and compromise quality in combined objective space.

This combined analysis strategy is justified because it matches the nature of the research problem. Descriptive statistics provide clarity, inferential tests provide statistical support, and Pareto analysis provides decision insight. Together, they allow the study to move from raw performance data to a defensible comparison and trade-off analysis, rather than to a single winner claim.

9 RESEARCH QUALITY: ETHICAL AND SAFETY ISSUES

9.1 Research quality

9.1.1 Reliability and Validity of the data collection instrument (if quantitative)

This integrated strategy supports positivist validity requirements by making claims measurable, replicable, and statistically testable, while producing scenario-conditioned decision guidance for practical controller placement decisions.

This study will ensure internal validity through strict control of simulation variables so that observed differences in performance can be attributed to the controller placement method rather than to uncontrolled experimental noise. The same topology families, scale levels, traffic assumptions, and evaluation procedures will be applied consistently across all experimental runs. Method comparisons will therefore be made under equivalent conditions, allowing a credible assessment of whether AI-driven methods genuinely outperform heuristic baselines.

Reliability will be strengthened through reproducibility controls. These include the use of fixed random seeds, standardised topology datasets such as Internet2- or GÉANT-type network structures, and repeatable simulation scripts. If the experiment is repeated under the same configuration, the results should be broadly consistent, subject to normal stochastic variation. This approach supports both repeatability and transparency, which are essential for a computational study of algorithmic performance.

9.1.2 Measures to ensure trustworthiness of data (if qualitative)

9.2 Ethical and safety considerations

9.2.1 Informed consent

This study does not involve human participants, survey respondents, or interviewees. It is an algorithmic evaluation conducted entirely within simulated network environments. As a result, traditional informed consent procedures do not apply.

9.2.2 Debriefing

Debriefing is not applicable because no human participants are exposed to the study procedures. The outputs of the study are simulation results, performance metrics, and comparative analyses of algorithms.

9.2.3 Withdrawal from investigation

Participant withdrawal does not apply because no participants are recruited. However, experimental runs may be stopped, repeated, or discarded if they are found to be corrupted, unstable, or methodologically invalid.

9.2.4 Protection of participants

The study is ethical because it is conducted in a sandboxed simulation environment, such as Mininet and Python-based experimental scripts, and therefore poses no risk to live university network infrastructure, operational systems, or human well-being. The work is limited to controlled computational experiments and does not interfere with production services or expose real users to harm.

9.2.5 Confidentiality and anonymity

Confidentiality concerns are minimal because the study will use only public or synthetic topology data and will not require access to sensitive institutional network configurations. No private network identifiers, user data, or operational logs from live systems will be collected or disclosed. This ensures that the research remains ethically appropriate and does not compromise institutional confidentiality.

9.2.6 Data storage

Research data, experimental logs, scripts, and generated outputs will be stored securely in password-protected local environments during active analysis. Backup copies will be maintained in encrypted cloud storage approved by the University, such as OneDrive or Google Drive where available. Version control will be managed through a private GitHub or Bitbucket repository to preserve a clear audit trail of code and analysis changes. Access to all files will be restricted to the researcher and, where required, the supervisor. This storage plan supports data integrity, traceability, and responsible research practice.

Overall, this study is ethical because it adheres to the principles of research integrity, transparency, and computational safety. It involves no human participants, does not expose sensitive systems, and is designed to generate reproducible evidence within a controlled and low-risk technical environment.

10 INTELLECTUAL PROPERTY

This research is conducted under the auspices of the University of Zululand, and all intellectual property generated during the study shall belong to the University in accordance with institutional policy. This includes the research design, analytical framework, experimental artefacts, scripts, and any original outputs produced for the project.

The source code developed for the ai-sdn-controller-placement project will be documented and managed in line with the University’s research integrity guidelines. All code contributions, simulation assets, and analytical outputs will be version-controlled, appropriately attributed, and retained in a manner that supports transparency, reproducibility, and academic accountability. Where external datasets, libraries, or tools are used, they will be acknowledged in accordance with applicable licensing and citation requirements.

11 RESOURCES REQUIRED AND PROJECT PLAN

11.1 Resources required

This study requires a modest set of technical resources because it is a software-based and simulation-driven investigation rather than a field study.

The primary resource is a reliable development machine, preferably a laptop or desktop with at least a quad-core processor, 16 GB RAM, and sufficient storage space for simulation logs, datasets, and version-controlled project files. A larger memory capacity is desirable because repeated simulation runs and analysis scripts may produce substantial log output. A stable internet connection is also required for literature access, package installation, and repository synchronization.

The software environment will consist of Python 3.x together with the libraries required to implement, run, and analyse the experiments. These include standard scientific and data-processing libraries such as NumPy, Pandas, SciPy, and Matplotlib, as well as simulation and network-modelling tools such as Mininet where applicable. If the experimental workflow includes custom orchestration or metric extraction, additional Python utilities may be used to support automation, parsing, and visualisation. For project management and reproducibility, the source code will be maintained in the ai-sdn-controller-placement repository using version control.

Cloud storage will also be used for secure backup and synchronisation of scripts, results, and analysis outputs. This ensures continuity of work, prevents accidental data loss, and supports traceability across the project lifecycle.

11.2 Budget

Budget (if applicable) is reasonable and matches the study? (Table format)

11.3 Research process - Gantt chart

(below is an example – please delete and develop your own)

Table 1: xxxxxx

No.	Milestones and Dates	May 2022	August 2022	September 2022	October 2022	November 2022	Jan - March 2023	April- May 2023	June - July 2023	August 2023	September 2023	October 2023	November 2023

1	*Proposal development*												
2	*Submit proposal*												
3	*Receive feedback and Corrections*												
4	*Submit a proposal for review*												
5	*Receive feedback and Corrections*												
6	*Present proposal to department /faculty*												
7	*HDC approval*												
8	*Thesis / Dissertation development*												
	- Further Literature review												
	- Conceptual framework												
	- Data collection												
	- Data analysis/ methodology												
	- Discussion, conclusion & recommendations												
	- Thesis submission for examination												

12 FEASIBILITY OF THE STUDY

The study is feasible because it is technically manageable, methodologically contained, and aligned with the researcher's current capabilities and resources. The researcher has a background in Computer Science and Mathematics, which provides the necessary foundation for understanding algorithmic design, simulation logic, performance evaluation, and statistical comparison. This background is appropriate for a study that combines network optimisation, experimental analysis, and quantitative reasoning.

The study is also feasible because the required resources are readily available. The technical environment is based on a personal development machine, Python, open-source libraries, and simulation tools such as Mininet, all of which can be used without specialised infrastructure. The project does not depend on human participants, physical network deployment, or costly proprietary software, which reduces both complexity and risk.

The study is further feasible because the timeline is realistic for an Honours submission. The work is divided into clearly defined stages covering review, setup, experimentation, analysis, and writing. This sequencing allows progress to be measured and adjusted if needed, while keeping the project within a single academic cycle.

The study is feasible because the scope is controlled, the methods are reproducible, the computational requirements are moderate, and the expected outputs are achievable within the available time frame. The project therefore satisfies the practical and academic conditions necessary for successful completion at Honours level.

13 DISSEMINATION PLAN

NB: See the attached guide for clues or hints on how to write the dissemination plan. Book, book chapters, conference, or papers to be published indicated? Access to post-study information given to community (if applicable)? Reciprocity and

adequacy: Is there a clear benefit shown for the university and the participating partners/ community? (how do we give back to the community engaged) (Is there a plan to give back?)

14 PROPOSED CHAPTER DIVISION

Chapter 1: Introduction and Background.

Chapter 2: Literature Review and Conceptual Framework.

Chapter 3: Research Methodology (Design, Instruments, and Procedures).

Chapter 4: Presentation of Results and Data Analysis.

Chapter 5: Summary, Conclusion, and Recommendations.

15 REFERENCES

(Use the APA or Harvard format) (Zotero is recommended)

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